

PROBLEM:

Given the O.D.E.
(CONTINUOUS TIME)

$$\gamma \frac{dx}{dt} = -x + f(t)$$

FIND A (DISCRETE TIME) "DIFFERENCE" EQUATION

$$\hat{X}_{k+1} = a \hat{X}_k + b \int_k^{\hat{}} \text{ SUCH THAT}$$

$$\hat{X}_k \sim X(k \cdot \Delta t) \quad \text{IN "SOME SENSE",}$$

(IF $f(t)$ IS A STOCHASTIC PROCESS)

1) EULER'S METHOD $\frac{dx}{dt} \sim \frac{x(t+\Delta t) - x(t)}{\Delta t}$

$$x(t+\Delta t) = x(t) + \frac{\Delta t}{\gamma} (-x(t) + f(t)) = \left(1 - \frac{\Delta t}{\gamma}\right) x(t) + \frac{\Delta t}{\gamma} f(t)$$

$$x(t+\Delta t) \approx x(t) + \frac{\Delta t}{\gamma} (-x(t) + f(t)) = \left(1 - \frac{\Delta t}{\gamma}\right) x(t) + \frac{\Delta t}{\gamma} f(t)$$

2) USE THE ANALYTICAL SOLUTION TO GET A "BETTER" WAY. "BETTER" MEANS "WITH SOME RATIONALE & NOT EMPIRICAL"

$$x(t) = \text{SOLUTION OF THE HOMOGENEOUS} + \text{CONVOLUTION INTEGRAL}$$

$$x(t) = x(t_0) \cdot h(t) + \int_{-\infty}^{+\infty} h(t-y) f(y) dy$$

$$h(t) = e^{-t/\tau} \cdot \Theta(t)$$

$$\Theta(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

HEAVISIDE
FUNCTION
(GRADINO)

Note: $\delta(t) = \Theta'(t)$

$$\int_{-\infty}^{+\infty} e^{-\frac{(t-y)}{\tau}} \cdot \Theta(t-y) f(y) dy =$$

$$= \underbrace{\int_{-\infty}^t e^{-\frac{t-y}{\tau}} \cdot \Theta(t-y) f(y) dy}_{y \in (-\infty; t)} + \underbrace{\int_t^{+\infty} e^{-\frac{t-y}{\tau}} \cdot \Theta(t-y) f(y) dy}_{y \in (t; +\infty)} =$$

$$\Leftrightarrow y < t \Leftrightarrow (t-y) > 0$$

$$\Rightarrow \Theta(t-y) = 1$$

$$y \in (t; +\infty)$$

$$\Leftrightarrow y > t \Leftrightarrow (t-y) < 0$$

$$\rightarrow \Theta(t-y) = 0$$

$$= \int_{-\infty}^t e^{-\frac{(t-y)}{\tau}} \cdot f(y) dy$$

$$x(t+\Delta t) = x(t_0) e^{-\frac{t+\Delta t}{\tau}} + \int_{-\infty}^{t+\Delta t} e^{-\frac{(t+\Delta t)-y}{\tau}} f(y) dy =$$

$$= x(t_0) e^{-t/\tau} \cdot e^{-\Delta t/\tau} + \int_{-\infty}^t e^{-\frac{(t-y)}{\tau}} \cdot e^{-\Delta t/\tau} f(y) dy + \int_t^{t+\Delta t} e^{-\frac{(t-y)}{\tau}} \cdot e^{-\Delta t/\tau} f(y) dy$$

$$= \underbrace{x(t_0) \cdot e^{-t/\tau} \cdot e^{-\Delta t/\tau} + e^{-\Delta t/\tau} \cdot \int_{-\infty}^t e^{-\frac{(t-y)}{\tau}} f(y) dy}_{e^{-\Delta t/\tau} \cdot x(t)} + e^{-\Delta t/\tau} \cdot \int_t^{t+\Delta t} e^{-\frac{(t-y)}{\tau}} f(y) dy$$

$$e^{-\Delta t/\tau} \int_t^{t+\Delta t} e^{-\frac{(t-y)}{\tau}} \cdot f(y) dy$$

↙ $f(t)$ IS DETERMINISTIC *

↘ $f(t)$ IS (CONTINUOUS) STOCHASTIC, ZERO MEAN UNITARY VARIANCE
 AUTOCORR $\delta(\tau)$
 i.e. $\langle f(a)f(b) \rangle = \delta(a-b)$ *

* When $\Delta t \ll 1$, $f(t) \approx \text{CONSTANT}$ IN $(t \text{ to } t+\Delta t)$, SO I TAKE IT OUT OF

$$\begin{aligned} &\approx e^{-\Delta t/\tau} \cdot f(t) \int_t^{t+\Delta t} e^{-\frac{(t-y)}{\tau}} dy = e^{-\Delta t/\tau} \cdot f(t) \cdot \tau \left. e^{-\frac{(t-y)}{\tau}} \right|_t^{t+\Delta t} \\ &= e^{-\Delta t/\tau} f(t) \cdot \tau \cdot (e^{\Delta t/\tau} - 1) \\ &= \tau f(t) \cdot (1 - e^{-\Delta t/\tau}) \end{aligned}$$

$$\Rightarrow x(t+\Delta t) \approx e^{-\Delta t/\tau} \cdot x(t) + \tau (1 - e^{-\Delta t/\tau}) \cdot f(t)$$

IF YOU USE TAYLOR, YOU RE-DISCOVER EULER'S METHOD

* $x(t+\Delta t) \approx e^{-\Delta t/\tau} \cdot x(t) + e^{-\Delta t/\tau} \cdot$

$$\langle \phi \rangle = \int_t^{t+\Delta t} e^{-\frac{(t-y)}{\tau}} \cdot \langle f(y) \rangle dy$$

$$\int_t^{t+\Delta t} e^{-\frac{(t-y)}{\tau}} \cdot f(y) dy$$

R.V. ! which properties?
 DISCRETE TIME

ϕ IS GAUSSIAN (LINEARITY)

$$\langle \phi^2 \rangle = \left\langle \left(\int_t^{t+\Delta t} e^{-\frac{(t-y)}{\tau}} \cdot f(y) dy \right) \left(\int_t^{t+\Delta t} e^{-\frac{(t-\epsilon)}{\tau}} \cdot f(\epsilon) d\epsilon \right) \right\rangle = \iint_t^{t+\Delta t} e^{-\frac{(t-y) + (t-\epsilon)}{\tau}} \cdot \langle f(y)f(\epsilon) \rangle dy d\epsilon$$

(3)

$$\langle f(y)g(\epsilon) \rangle = \delta(y-\epsilon) \rightarrow \iint e^{-\frac{(2t-y-\epsilon)}{\tau}} \cdot \delta(y-\epsilon) dy d\epsilon$$

$$= \int_t^{t+\Delta t} e^{-\frac{2(t-y)}{\tau}} dy =$$

$$= \tau e^{2\Delta t/\tau}$$

↑
MAKES ONLY
THE $y=\epsilon$
"SURVIVE" THE
INTEGRAL

$$\Rightarrow x(t+\Delta t) \approx e^{-\Delta t/\tau} \cdot x(t) + \text{R.V.} \cdot e^{-\Delta t/\tau} \cdot \phi$$

R.V.
GAUSSIAN

SAY THAT I CALL "m" A
GAUSSIAN R.V. WITH ZERO
MEAN
AND UNITARY
VARIANCE

$$\phi = \sqrt{\tau e^{2\Delta t/\tau}} \cdot m$$

← This is a known property of GAUSSIAN R.V.
1) ADDING m TO IT, MAKES THE MEAN m
2) MULTIPLYING BY S, MAKES THE VAR S^2